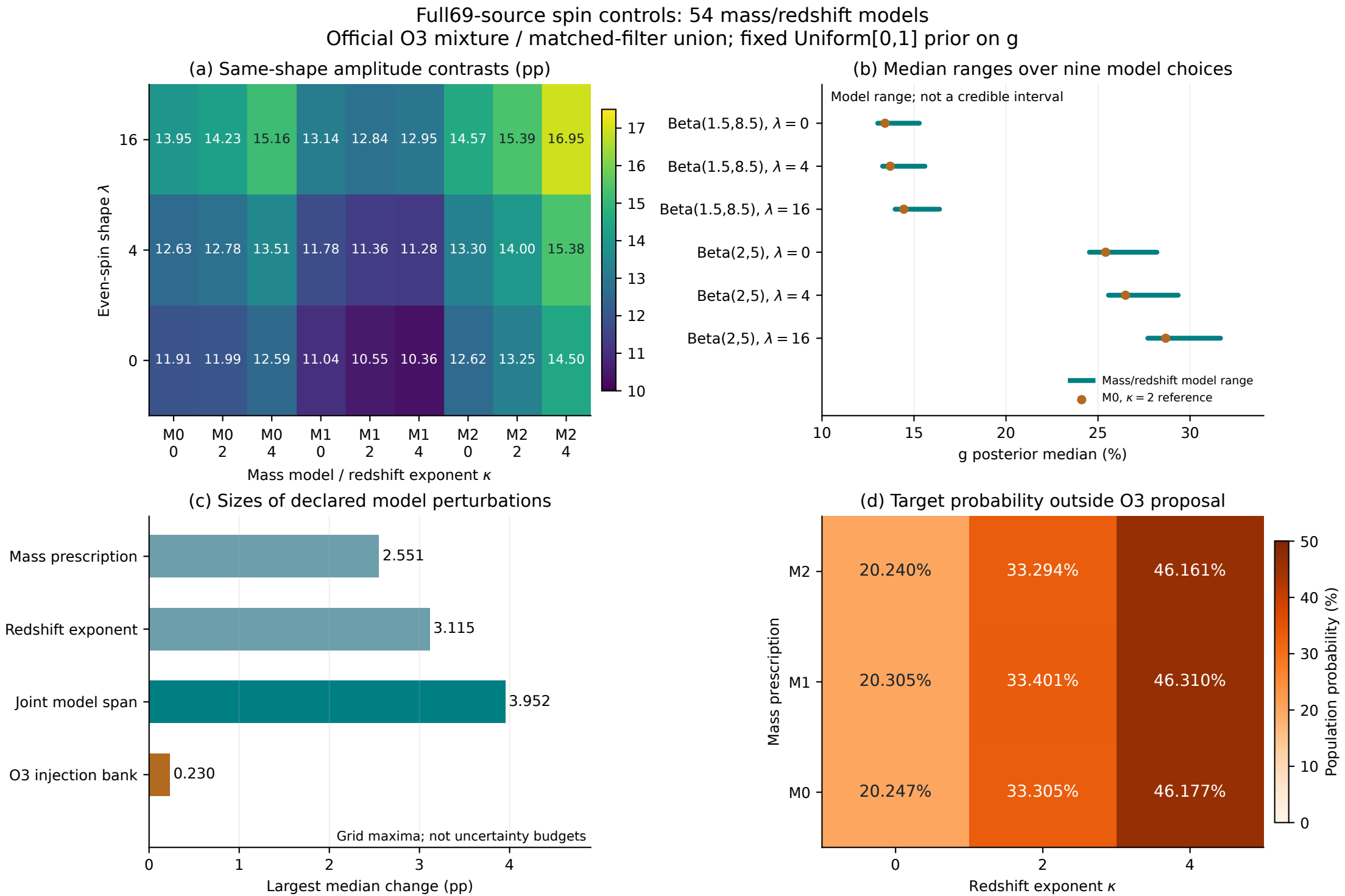

Full69-source spin controls: 54 mass/redshift models
Official O3 mixture / matched-filter union; fixed Uniform[0,1] prior on g
(a) Same-shape amplitude contrasts (pp)
Even-spin shape λ
16
4
0
13.95 14.23 15.16 13.14 12.84 12.95 14.57 15.39 16.95
12.63 12.78 13.51 11.78 11.36 11.28 13.30 14.00 15.38
11.91 11.99 12.59 11.04 10.55 10.36 12.62 13.25 14.50
17
16
15
14
13
12
11
10
M0 0
M0 2
M0 4
M1 0
M1 2
M1 4
M2 0
M2 2
M2 4
Mass model / redshift exponent κ
(b) Median ranges over nine model choices
Model range; not a credible interval
Beta(1.5,8.5), λ = 0
Beta(1.5,8.5), λ = 4
Beta(1.5,8.5), λ = 16
Beta(2,5), λ = 0
Beta(2,5), λ = 4
Beta(2,5), λ = 16
Mass/redshift model range
M0, κ = 2 reference
10
15
20
25
30
g posterior median (%)
(c) Sizes of declared model perturbations
Mass prescription
Redshift exponent
Joint model span
O3 injection bank
2.551
3.115
3.952
0.230
Grid maxima; not uncertainty budgets
0
1
2
3
4
Largest median change (pp)
(d) Target probability outside O3 proposal
Mass prescription
M2
M1
M0
20.240% 33.294% 46.161%
20.305% 33.401% 46.310%
20.247% 33.305% 46.177%
0
2
4
Redshift exponent κ
50
40
30
20
10
0
Population probability (%)
M0: α = 2 + 10% peak; M1: α = 3 + 10% peak; M2: α = 2, no peak. Missing population mass is not detection error.